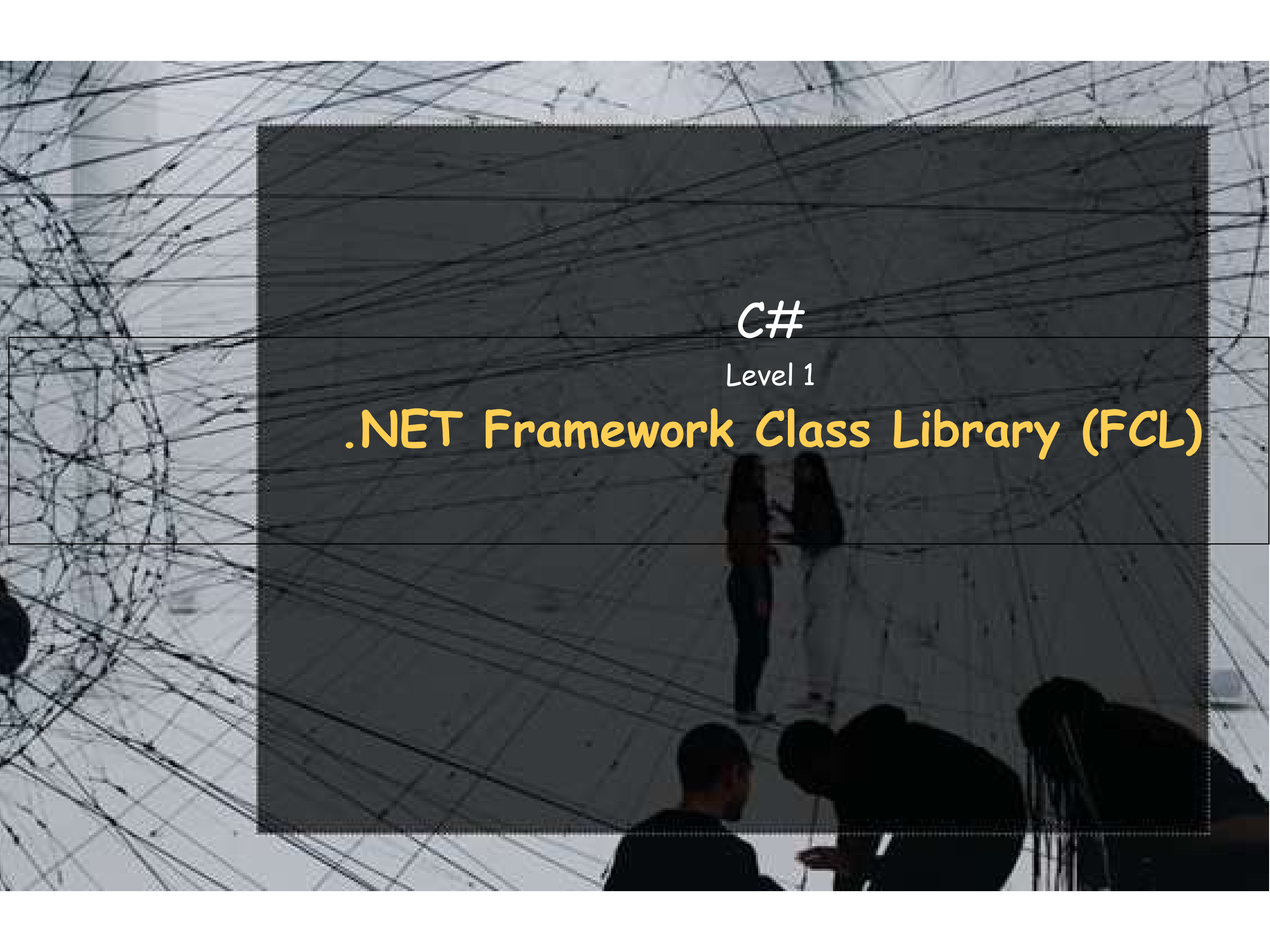


The background is a blurred image of a computer monitor. It shows a dark-themed code editor with various lines of code in different colors (blue, green, yellow). At the bottom, a taskbar is visible with several application icons, including what looks like a web browser, a file explorer, and some development tools. The overall image is out of focus, serving as a backdrop for the text.

C#

Level 1

.NET Framework Class
Library (FCL)



C#

Level 1

.NET Framework Class Library (FCL)

Definition

FCL

 FCL (Framework Class Library) is the collection of reusable classes, interfaces, types, and namespaces provided by the classic .NET Framework.

 It gives developers ready-to-use functionality instead of requiring everything to be built from scratch.

Explanation

Common Functionality

-  Data types and strings.
-  Collections.
-  File and stream operations.
-  Database access.
-  Networking.
-  XML processing.
-  Windows Forms.
-  Web functionality in classic .NET Framework.

BCL

 BCL (Base Class Library) represents the core/fundamental part of the .NET class libraries.

 In classic .NET Framework terminology, FCL is the broader library concept that includes the BCL.

Procedures

1. Choose the functionality needed by the application.
2. Reference the required namespace/library.
3. Use the provided classes and APIs.
4. Build the application without implementing common functionality from scratch.

💡 Example

```
using System.IO;  
File.WriteAllText("data.txt", "Hello");
```

➡ Result

The application uses existing .NET functionality to write data to a file.

Note

- ◆ Common namespaces include ``System``, ``System.Collections``, ``System.IO``, ``System.Net``, ``System.Data``, ``System.Text``, ``System.Threading``, and ``System.Reflection``.
- ◆ ``System.Web`` belongs to classic .NET Framework and is not part of modern ASP.NET Core.

⊘ Warning

- ◆ FCL and CLR are not the same.
- ◆ FCL provides libraries and APIs; CLR provides the runtime environment.
- ◆ Some FCL technologies belong specifically to the older .NET Framework.



Comparison



CLR

- ✓ Executes and manages .NET applications.
- ✓ Provides runtime services.
- ✓ Includes JIT and Garbage Collection.



FCL

- ✓ Provides reusable classes and APIs.
- ✓ Provides functionality for applications.
- ✓ Includes areas such as I/O, collections, data access, and networking.

✦✦ Summary

- ✓ FCL = .NET Framework Class Library.
- ✓ Provides reusable functionality.
- ✓ Includes the core BCL plus broader framework libraries.
- ✓ Supports files, collections, data, networking, XML, and more.
- ✓ CLR runs the application.
- ✓ FCL provides tools and functionality used by the application.

Thank you 😊